

Using Statistical Tools to Identify Poor Behavior or Fraud

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Our Solution:

Add Another Reviewer

Always Available. Lightning Fast Response.



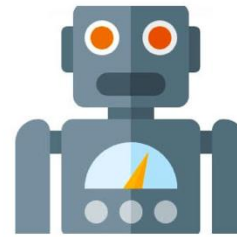
Reviewer 1



Reviewer 2



Reviewer 3



Reviewer 4

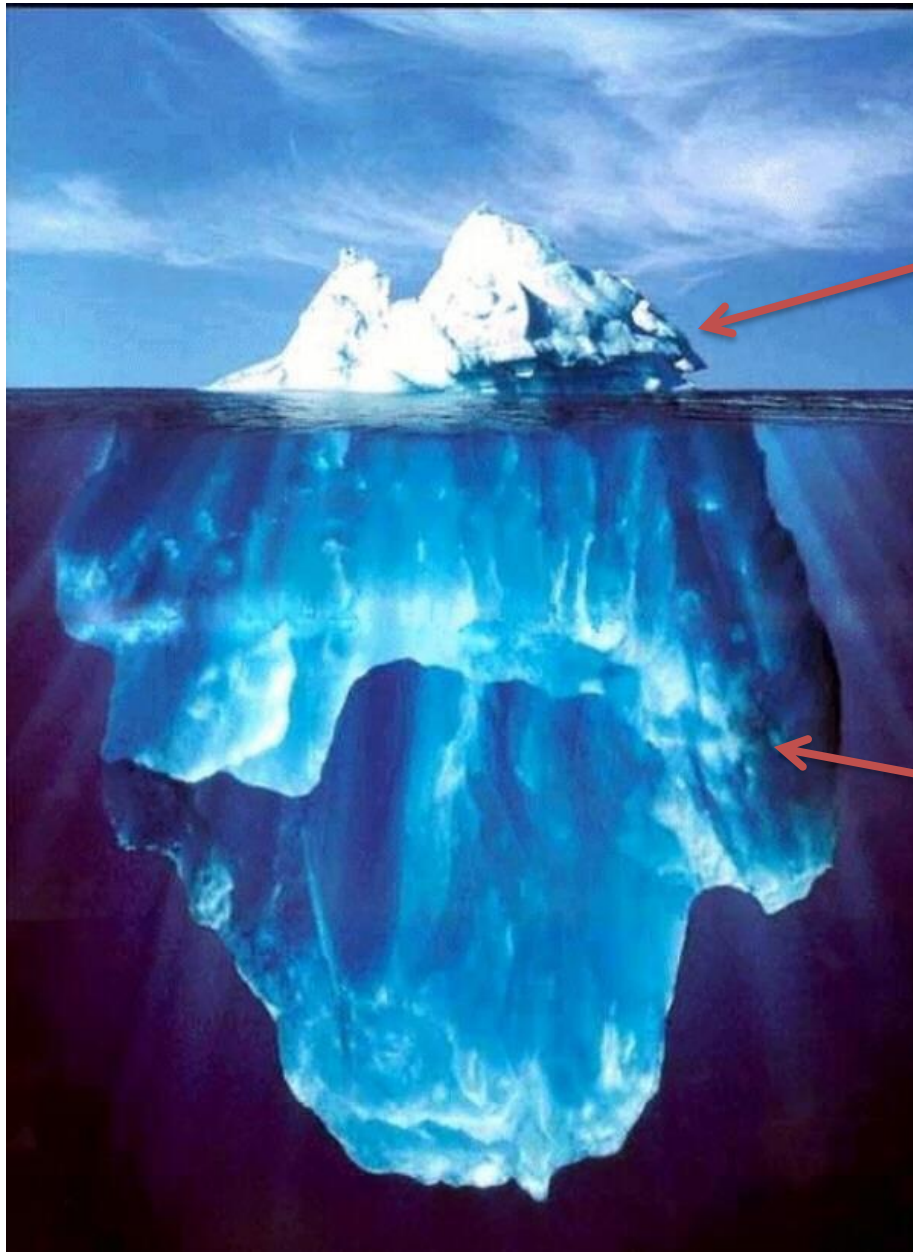
↑ Hint: This is StatReviewer

StatReviewer Content

- *Thousands* of algorithms are run on every manuscript. We check:
 - obvious numerical errors
 - quality of reporting (e.g., defined descriptive statistics)
 - appropriate statistical tests (e.g., t-test for skewed data)
 - style (e.g., precision of decimal places)
 - Methodological reporting

Fraud in Anesthesiology

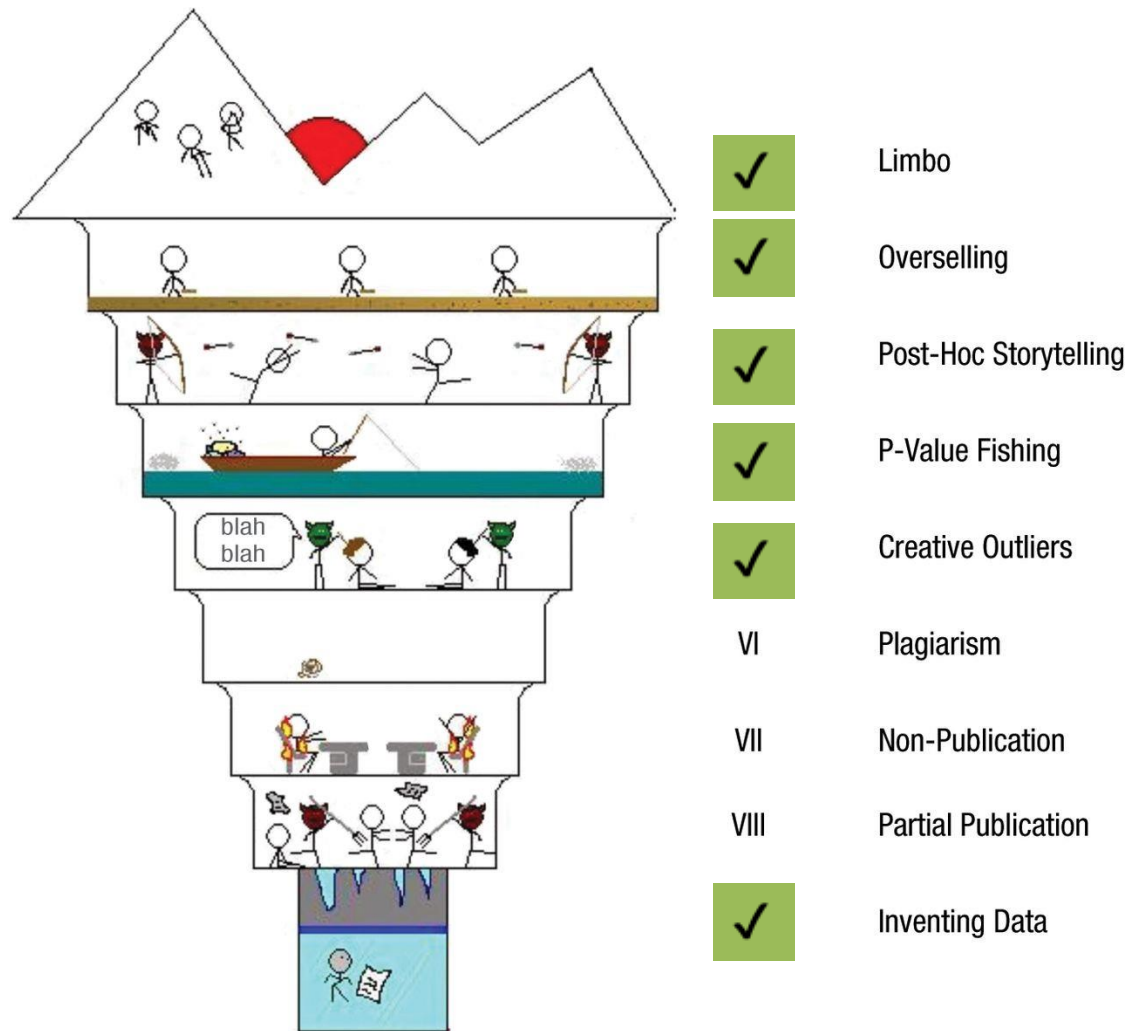
- Scott Reuben, MD (1996 to 2010)
 - 21 fraudulent scientific articles retracted
 - Admits that he never conducted any of the trials he reported in the literature
 - Sentenced to prison (2010)
- Joachim Boldt, MD (1999 to 2012)
 - False data in 88 (of 102) scientific articles retracted
- Yoshitaka Fujii, MD (1993 to 2012)
 - 172 (of 212) fraudulent scientific articles, with 126 articles entirely fabricated
 - No ethical oversight, authorship inclusions



Data Fabrication/Falsification

Unethical research behavior
(Conscious or Unconscious)

Fig. 1. The nine circles of scientific hell (with apologies to Dante and xkcd).



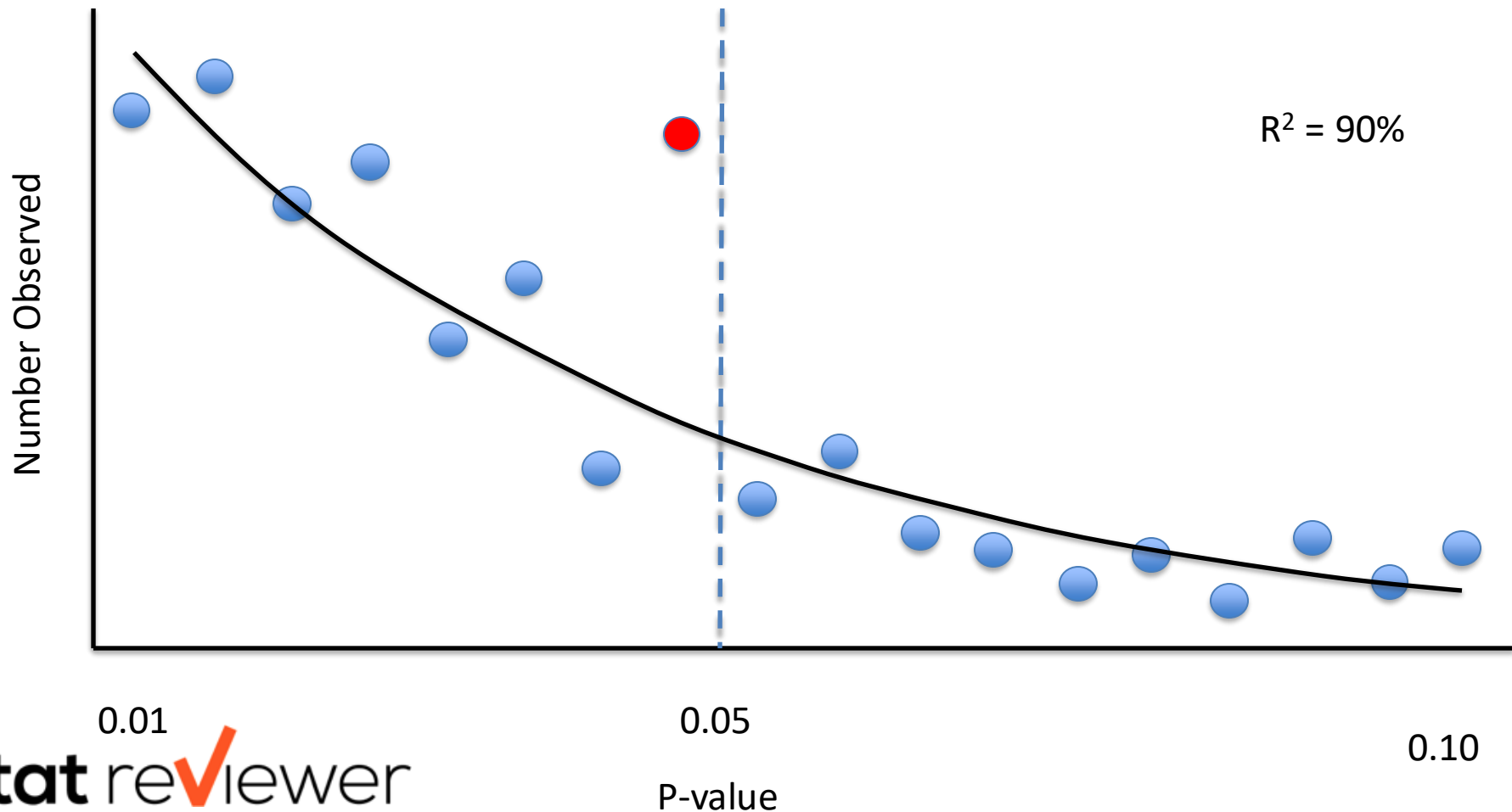
Neuroskeptic Perspectives on Psychological Science 2012;7:643-644

Perspectives on
Psychological
SCIENCE

Post hoc Story Telling: Forking Paths

- One reasonable hypothesis maps on to several reasonable statistical hypotheses (i.e., one to many)
 - PONV is associated with age
- Statistical interaction (moderation) must be taken into consideration for the primary interpretation
 - Age x sex is needed to consider the effect of age
- Adding confounder control post hoc
 - We should control for several covariates as they seem to be confounding the age association

A Peculiar Prevalence of P-Values Just Below 0.05: Masicampo & Lalande (2012)



Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

Boutron I, Dutton S, Ravaud P, Altman D. (May, 2010)

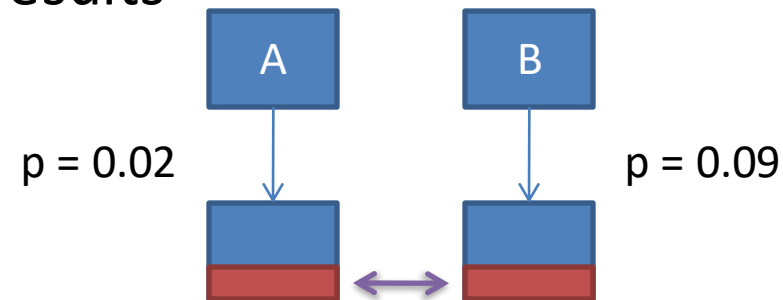
- Frequency and degree of “Spin” in published RCTs
- Spin[†]: specific reporting that could distort the interpretation of results and mislead readers.
 - From ignorance of scientific (statistical) issue
 - Unconscious bias
 - Willful intent to deceive

Reporting and Interpretation of Randomized Controlled Trials With Statistically Nonsignificant Results for Primary Outcomes

Boutron I, Dutton S, Ravaud P, Altman D. (May, 2010)

Strategies of Spin

- A focus on statistically significant results
 - within-group comparison
 - secondary outcomes
 - subgroup analyses
 - modified population of analyses



- Interpreting statistically nonsignificant results for the primary outcomes as showing treatment equivalence or comparable effectiveness
- Claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results.

Migraine Intervention With STARFlex Technology (MIST) Trial

A Prospective, Multicenter, Double-Blind, Sham-Controlled Trial to Evaluate the Effectiveness of Patent Foramen Ovale Closure With STARFlex Septal Repair Implant to Resolve Refractory Migraine Headache

Andrew Dowson, MBBS, PhD; Michael J. Mullen, MBBS, MRCP, MD; Richard Peatfield, MD, FRCP; Keith Muir, MD, FRCP; Arif Anis Khan, MBBS, FCPS; Christopher Wells, MB, ChB, FRCA; Susan L. Lipscombe, MB, ChB, MRCP; Trevor Rees, MB, ChB; Joseph V. De Giovanni, MD, FRCP, FRCPCH, MOM; W. Lindsay Morrison, MD, FRCP; David Hildick-Smith, MD, FRCP; Giles Eltrington, MD; W. Stewart Hillis, MB, ChB, FRCP, FRCS; Iqbal S. Malik, MA, MRCP, PhD; Anthony Rickards, MBBS, FRCP, FESC†

Background—Patent foramen ovale (PFO) is prevalent in patients with migraine with aura. Observational studies show that PFO closure resulted in migraine cessation or improvement in ~80% of such patients. We investigated the effects of PFO closure for migraine in a randomized, double-blind, sham-controlled trial.

Methods and Results—Patients who suffered from migraine with aura, experienced frequent migraine attacks, had previously failed ≥ 2 classes of prophylactic treatments, and had moderate or large right-to-left shunts consistent with the presence of a PFO were randomized to transcatheter PFO closure with the STARFlex implant or to a sham procedure. Patients were followed up for 6 months. The primary efficacy end point was cessation of migraine headache 91 to 180 days after the procedure. In total, 100 of 432 patients (38%) had right-to-left shunts consistent with a moderate or large PFO. One hundred forty-seven patients were randomized. No significant difference was observed in the primary end point of migraine headache cessation between implant and sham groups (3 of 74 versus 3 of 73, respectively; $P=0.51$). Secondary end points also were not achieved. On exploratory analysis, excluding 2 outliers, the implant group demonstrated a greater reduction in total migraine headache days ($P=0.027$). As expected, the implant arm experienced more procedural serious adverse events. All events were transient.

Conclusions—This trial confirmed the high prevalence of right-to-left shunts in patients with migraine with aura. Although no significant effect was found for primary or secondary end points, the exploratory analysis supports further investigation. The robust design of this study has served as the model for larger trials that are currently underway in the United States and Europe. (*Circulation*. 2008;117:1397-1404.)

Detecting Fabricated Data in RCTs

- J.B. Carlisle (2012)* applied a clever strategy to examine the probability that Fujii's data were the result of non-random processes
 - Distribution of patient characteristics “incredibly nice” (Kranke et al, 2000)
 - Patients in randomized groups *too similar* to each other
- Examined 168 of Fujii RCTs
 - Compared expected distributions to observed distributions of patient characteristics

The “Table 1” of an RCT

Table 1. Patient and Labor Characteristics of Laboring Patients

	Study group (neostigmine + bupivacaine) (<i>n</i> = 20)	Control group (bupivacaine alone) (<i>n</i> = 20)
Age (yr)	27 ± 6	29 ± 7
Height (cm)	166 ± 5	164 ± 9
Weight (kg)	84 ± 14	84 ± 15
Gestation (wk)	39 ± 1	39 ± 1
Parous (%)	45%	45%
Oxytocin use (%)	55%	75%
Oxytocin infusion rate (mU/min) at time of epidural catheter placement ^a	7.0 ± 10.3	8.4 ± 7.6
Fetal weight (g)	3350 ± 540	3370 ± 450
VRS score-baseline ^b	7.6 ± 1.8	6.9 ± 2.2
VRS score-0 min ^c		
Mean ± SD	0.6 ± 1.3	0.8 ± 1.3
Median	0	0
Mode	0	0
Range	0–3	0–3
Cervical dilation prior epidural placement (cm)	3.1 ± 1.3	3.3 ± 1.3
Cervical dilation 2 h after epidural placement (cm)	4.7 ± 1.5	4.3 ± 1.5

Data are presented as mean ± SD or incidence (%) unless otherwise indicated. There are no statistical differences between groups for any variable.

VRS = verbal rating scale.

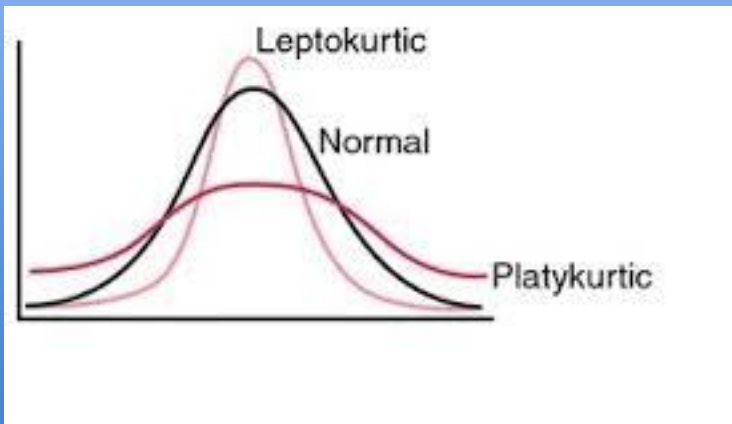
^a Average of all patients including those with or without oxytocin use at time of epidural catheter placement.

^b VRS before initiation of epidural analgesia.

^c Twenty minute after injection of epidural study solution.

The “Table 1” of an RCT

The standardized difference between groups should be normally distributed with A mean of 0.

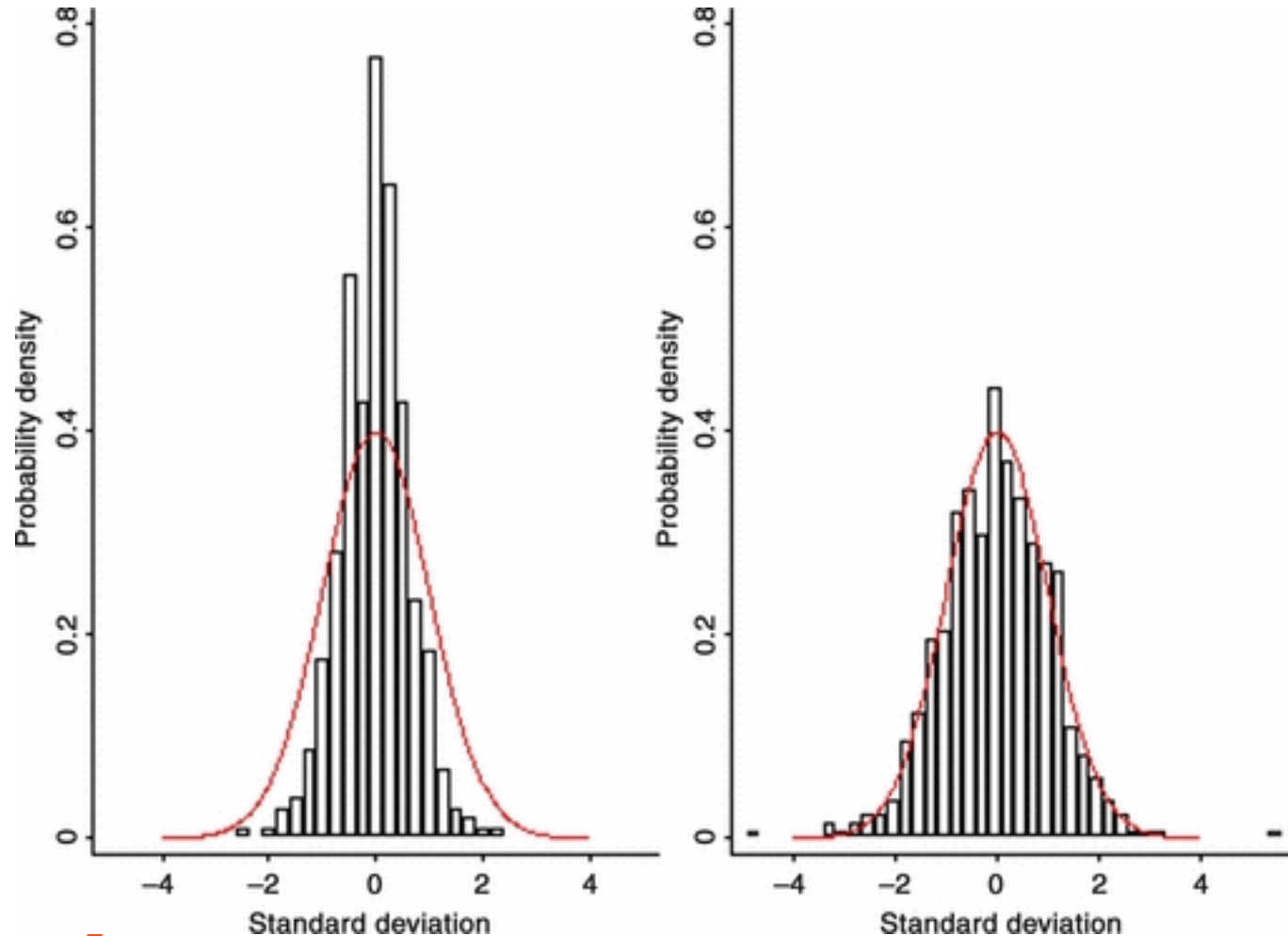


Study group (neostigmine + bupivacaine) (n = 20)	Control group (bupivacaine alone) (n = 20)
27 ± 6	29 ± 7
166 ± 5	164 ± 9
84 ± 14	84 ± 15
39 ± 1	39 ± 1
45%	45%
55%	75%
7.0 ± 10.3	8.4 ± 7.6
3350 ± 540	3370 ± 450
7.6 ± 1.8	6.9 ± 2.2
0.6 ± 1.3	0.8 ± 1.3
0	0
0	0
0-3	0-3
3.1 ± 1.3	3.3 ± 1.3
4.7 ± 1.5	4.3 ± 1.5

no statistical differences between groups for any variable.

atheter placement.

Fujii (left) versus others (right) on Age



Thank You!

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